60 MeV). By using two separate tanks and alternatingly positively and negatively charged ions, three-stage VdG machines with maximum proton energies up to ~45 MeV have been built.

The energy of a beam produced by a VdG generator is extremely precise. However, the current intensity, 10–100 μA, is somewhat less than that of other accelerators. The beam current i (amperes) is given by the relation

$$i = q\, I_o = \mathbf{e}\, z\, I_0 \tag{16.4}$$

where q is the projectile charge (C), I_0 the incident particle current (particles s^{-1}), $\mathbf{e} = 1.6 \times 10^{-19}$ C, and z the net charge (in electron units) of the beam particle (ion).

16.5. MULTIPLE-STAGE LINEAR ACCELERATORS

The potential obtained from a high voltage generator can be used repeatedly in a multiple-stage accelerator process ($n > 1$ in (16.1)). The linear accelerator operates on this principle. The accelerator tube consists of a series of cylindrical electrodes called drift tubes. In the Wideröe linear accelerator, the electrodes are coupled to a radio frequency generator in the manner shown in Figure 16.4. The high voltage generator gives a maximum voltage V which is applied to the electrodes by the radio frequency so that the electrodes alternate in the sign of the voltage at a constant high frequency. If the particles arrive at the gap between electrodes in proper phase with the radio frequency such that the exit electrode has the same charge sign as the particle and the entrance electrode the opposite, the particles are accelerated across the gap. Each time the particles are accelerated at an electrode gap they receive an increase in energy of qV; for n electrode gaps the total energy acquired is nqV (16.1). Inside the drift tubes no acceleration takes place since the particles are in a region of equal potential.

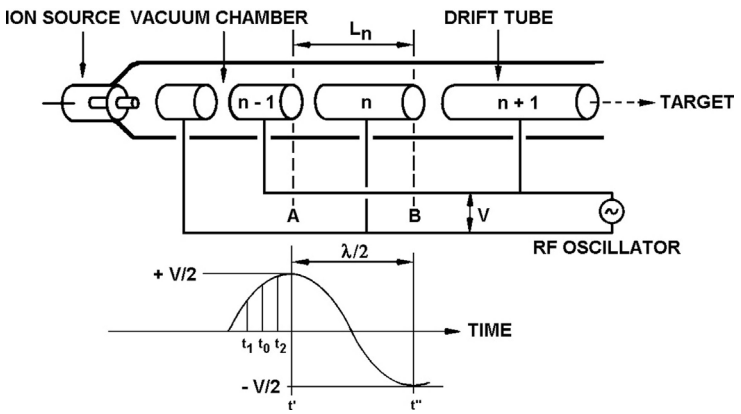


Figure 16.4 Principle of a multi-stage high voltage linear accelerator of Wideröe type.

Veksler and McMillan independently demonstrated the principle of phase stability, which is based on an effect which tends to keep the particles in phase with the radio frequency oscillation of the potential and allows beams of sufficient intensity for use in research. In the lower part of Figure 16.4 the variation of potential at point A with time is shown. In operation, a particle in proper phase should arrive at the acceleration gap at time t_o. If a particle has less energy, it takes longer time to traverse the drift tube and arrives late at the gap, e.g. at time t_2. Since the potential is now higher, it receives a greater acceleration, thereby becoming more in phase. If a particle is too energetic, it arrives too early at the gap t_1 and receives a smaller acceleration, thereby putting it better in phase. This bunching of the ions not only keeps the ions in phase for acceleration but also reduces the energy spread in the beam. When the velocity approaches **c**, phase stability vanishes gradually because changes in energy now mostly becomes changes in mass and not in velocity.

Since the velocity of the beam particles are increasing as they progress down the accelerator but the oscillator frequency remains constant, it is necessary to make the drift tubes progressively longer so as to allow the beam particles to arrive at the exit of each tube in phase with the oscillation of the potential. Acceleration to high energies requires relatively long acceleration tubes. The electric field between the tubes which accelerates the particles also spreads the beam. The defocussing effect can be minimized by the use of electrostatic and magnetic lenses, in the drift tubes, which refocus the beam onto the center line of the acceleration tubes.

For protons and deuterons the increase in energy below 10 MeV is directly related to the increase in the velocity of the particles. However, as the kinetic energy exceeds 10−20 MeV the relativistic mass becomes important since the speed is approaching the velocity of light (Fig. 5.2). At projectile energies above 100 MeV, the relativistic mass increase is greater than the increase in velocity and at very high energies, the GeV range, the energy increase is practically all reflected by an increase in the relativistic mass (at 1.9 GeV, the proton mass is double that of the rest mass).

The phase requirement (Fig. 16.4) is that the time from point A to B (i.e. t''-t') must be

$$L_n/v_n = \lambda/(2\mathbf{c}) \tag{16.5}$$

Taking the relativistic mass increase into consideration it can then be shown that the length of the drift tube at the nth stage is

$$L_n = (\lambda/2)\left[1 - (nk+1)^{-2}\right]^{1/2} \tag{16.6}$$

where

$$k = qV/\left(m^o\mathbf{c}^2\right) = z\mathbf{e}V/\left(m^o\mathbf{c}^2\right) \tag{16.7}$$

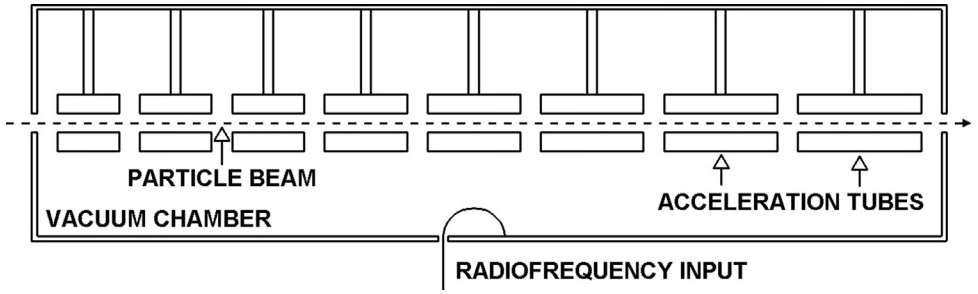


Figure 16.5 Alvarez type of linear accelerator.

When $v \rightarrow \mathbf{c}$, nk becomes $\gg 1$ and thus $L_n \rightarrow \lambda/2$. The length of the drift tubes therefore becomes constant at high particle energies.

In the Alvarez linear accelerator the drift tubes are mounted in a tank acting as a micro-wave resonator, see Figure 16.5. By varying the diameter with drift tube length it is possible to achieve resonance for all drift tubes. A powerful high-frequency radio signal is introduced into the tank by a small antenna. The standing electromagnetic wave introduces oscillating potential differences between the drift tubes in a way similar to the Wideröe accelerator. The drift tubes are grounded in the tank wall through their supports and no insulators are required. This permits higher potential differences between the tubes than in the Wideröe design.

A linear accelerator in use at the Gesellschaft für Schwerionenforschung at Darmstadt, Germany, is the UNILAC (for UNIversal Linear ACcelerator), which initially had a Wideröe section (beam energy 1.4 MeV/u) followed by Alvarez accelerators (beam energy 11.4 MeV/u) and 17 single step linear accelerator cavities. To increase the achievable energy, the partly accelerated ions are further stripped of electrons by passage through gas or nickel foil strippers located after the Wideröe section and after the Alvarez tanks, cf. (16.2). The machine was originally designed for synthesis of transuraniums and super-heavy elements through bombardment of targets of high atomic number elements with heavy ions. The maximum beam current is $\sim 10^{12}$ particles s^{-1} and the energy ≤ 20 MeV/u, which corresponds to ≤ 5 GeV for accelerated uranium ions. The UNILAC has been used to produce a number of transactinide elements (Ns, Hs, Mt). At present the Wideröe section has been replaced by a radio-frequency quadrupole accelerator (see below) followed by two drift tube linear accelerators. This has resulted in a higher ion current than before.

The world's largest linear accelerator is located at Stanford University in California. In this accelerator electrons and positrons are raised to energies of 50 GeV and used to study elementary particles, the distribution of charges within nuclei, etc. The accelerator tube is 3200 m long; the diameter is about 0.1 m. The maximum beam intensity is 8 pA. At 50 GeV the electrons have a velocity of 0.999 999 999 95 **c** and the

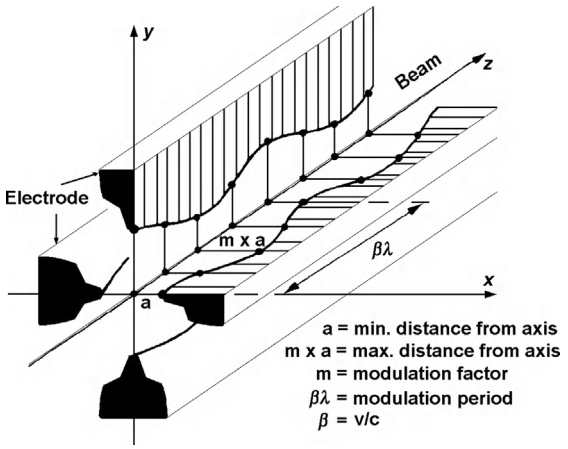


Figure 16.6 RFQ accelerator showing the four shaped electrodes (vanes) of the RF cavity.

electron mass is ~97 800 times greater than its rest mass (a little heavier than a chromium atom).

An important advance in linear accelerator technology is the development of radio-frequency quadrupoles (RFQ:s), permitting either simultaneous acceleration and focusing of high currents of particles or the funnelling of two or more beams into one which is then fed into a linear accelerator operating at a higher frequency. In principle the RFQ is somewhat similar to the Alvarez design but without separate drift tubes, see Figure 16.6. Its main characteristics can be summarized as: i) it is linear accelerator operating in the energy range 20 keV/u to 2 MeV/u, ii) it bunches, focuses, and accelerates particles with RF fields, iii) typical lengths are 1 to 2 m and operating frequencies 50 to 400 MHz, iv) intervane voltage ~100 kV, v) minimum value of *a* (see Figure 16.6) 3–4 mm, Q-factor ~10,000, vi) intervane capacity ~10^{10} F/m. In order to accept an incoming unbunched beam the a-value has to be rather high at the entrance and then progressively decrease to its final constant value. The undulating vane profiles are normally obtained by using numerically controlled milling machines.

16.6. CYCLOTRONS

The difficulties inherent in the length of high energy linear accelerators were ingeniously overcome by Lawrence and Livingston in their construction of the first cyclotron in 1931. The basic operating principles of the cyclotron are shown in Figure 16.7. The particles are accelerated in spiral paths inside two semicircular flat metallic cylinders called dees (D's), placed in a flat vacuum chamber. The two dees are connected to a high frequency alternating voltage. The volume within the dees corresponds to an equipotential condition just as does the volume within the drift tubes of linear accelerators. The